

Example 6: Find the entries in the second row of AB , where

$$A = \begin{bmatrix} 2 & -5 & 0 \\ -1 & 3 & -4 \\ 6 & -8 & -7 \\ -3 & 0 & 9 \end{bmatrix}, B = \begin{bmatrix} 4 & -6 \\ 7 & 1 \\ 3 & 2 \end{bmatrix}$$

Solution: $A_{21} = (-1) \cdot 4 + 3 \cdot 7 + (-4) \cdot 3 = 5$ (Row-Column Rule)

$$A_{22} = (-1) \cdot (-6) + 3 \cdot 1 + (-4) \cdot 2 = 1$$

$\therefore$ The second row is $[5 \ 1]$.

This observation about rows of AB is true in general and follows from the row-column rule.

Let $\text{row}_i(A)$ denote the i th row of a matrix A . Then

$$\boxed{\text{row}_i(AB) = \text{row}_i(A) \cdot B}$$

Properties of Matrix Multiplication

I_n represents the $n \times n$ identity matrix and $I_n x = x$ for all x in $\mathbb{R}^n$.

Theorem 2: Let A be an $m \times n$ matrix, and let B and C have sizes for which the indicated sums and products are defined.

a. $A(BC) = (AB)C$ (associative law of multiplication)

b. $A(B+C) = AB+AC$ (left distributive law)

c. $(B+C)A = BA+CA$ (right distributive law)

d. $r(AB) = (rA)B = A(rB)$

for any scalar r

e. $I_n A = A = A I_n$ (identity for matrix multiplication)